

```
results = model.fit()
print("Linear OLS ", results.params)
```

The application of the WLS algorithm is straightforward. The `sm.wls` function is used instead of `sm.ols`. It is necessary to pass the variable `w` containing the weights.

```
model = sm.wls(formula, df_data, weights=W)
results = model.fit()
print("Linear WLS ", results.params)
```

The computation of the quadratic calibration curve is linear except for the formula that becomes `formula = 'y ~ x + x2'`. The subsequent steps are the same.

```
formula = 'y ~ x + x2'
model = sm.ols(formula, df_data) # apply OLS algorithm
results = model.fit()
print("Polynomial OLS ", results.params)
model = sm.wls(formula, df_data, weights=W) # apply WLS
algorithm
results = model.fit()
print("Polynomial WLS ", results.params)
```

Table 2.3 summarizes the statistics returned by these few lines of code. The last column, called for Akaike information criterion (*AIC*) is explained further.

The differences between the coefficients calculated using the OLS or WLS algorithm appear small, except for the quadratic model. As previously stated, the coefficient of determination  $r^2$  does not bring much information. If the coefficients of the WLS quadratic model are applied to compute the inverse-predicted concentration of the samples of Table 2.2, the following results are obtained: 0.077, 0.074, 0.114, 0.113, 0.534, 0.543, 1.144, 1.113, 2.560, 2.486, 10.424, 10.829. The average bias remains at +15%. It is not very substantial to decide which model is the best. Another method is needed to select the best model, and a proposal based on MU is made in Section 8.1.

**Table 2.3** THEOPHYLLINE series 1 – comparison of results obtained with OLS and WLS methods on the data of the first series.

| Model     | Method | $a_0$  | $a_1$  | $a_2$   | $r^2$ | <i>AIC</i> |
|-----------|--------|--------|--------|---------|-------|------------|
| Linear    | OLS    | 0.5801 | 12.634 |         | 0.995 | 57.776     |
|           | WLS    | 0.0863 | 14.722 |         | 0.942 | 40.036     |
| Quadratic | OLS    | 1.0403 | 11.779 | 0.0837  | 0.995 | 59.496     |
|           | WLS    | 0.0489 | 16.312 | -0.4281 | 0.955 | 39.463     |

## 2.4 Calibration: Special Topics

### 2.4.1 Nonlinear Calibration Curve

For statisticians, the linearity of a model is not defined the same way as in other fields of mathematics. A model is “linear in the parameters” when it is an additive polynomial or can be transformed into such a polynomial, whatever the highest power of any explicative variable. For example, the quadratic model  $Y = a_0 + a_1X + a_2X^2$  used before is linear because the second order  $X^2$  variable can be transformed into a new first-order dummy variable  $X_2$ , containing the squared values of  $X$ . The model becomes  $Y = a_0 + a_1X + a_2X_2$ , and the three coefficients can be favorably estimated using `LINEST`.

The straight-line model is remarkably successful in analytical sciences because the coefficients are easily interpretable. If the sensitivity of a method is “the increase in response relative to the increase in concentration,” the coefficient  $a_1$  can be interpreted as the sensitivity.

Sensitivity

$$\lim_{\delta X \rightarrow 0} \frac{\delta Y}{\delta X} = a_1 \quad (2.33)$$

On the other hand, the coefficient  $a_0$  can be assimilated into a blank. Depending on the quantification technique, the *blank* can combine one or several following elements:

- The eventual response of the reagents used for the sample preparation.
- The *endogenous* analyte concentration of the sample when the matrix is present.
- The background instrumental response.

For many laboratory instruments, there is also an electrical blank (or auto zero) which allows the display to be reset to zero. When the matrix or the calibrant are absent and when the instrument is stable or does not present any drift, it is possible to confuse the calibration blank and the electrical blank. However, they do not have the same origin. The electrical zero is purely related to the electronics of the instrument, while the calibration blank may be produced by the reagents.

The situation is more complex if calibration is achieved in the presence of matrix containing an endogenous concentration of the analyte, the coefficient  $a_0$  is a mixture of instrumental, reagent, and sample responses as explained in Section 1.4. For some analytical techniques, it is also established that it is impossible to have a linear calibration model. In fact, it happens for numerous detection modes when the calibration extends over a wide concentration range.

For example, ICP emission spectrophotometry allows for very wide calibration ranges (three to four orders of magnitude) and often does not have a straight line for calibration curve. In this case, the interpretation of the coefficients is more delicate since the concept of sensitivity, as defined by the Eq. (2.33), is no longer meaningful: the sensitivity varies with the concentration.

For a given method, it is sometimes possible to modify the measured response and concentration to linearize the calibration model, but this modifies the statistical

properties of  $X$  and  $Y$  variables. For instance, an old practice was to linearize by using variable  $X$  and  $Y$  transformations, such as probit or log-probit transformation. This procedure is misleading and has major downsides, whereas it modifies the probability function of variables  $X$  and  $Y$ , which greatly degrades the inverse prediction accuracy and increases the MU. It is better to select a nonlinear model and keep it in its original form. This means it is necessary to apply a nonlinear regression method to estimate the coefficients, even if it seems more complicated.

The chemical mechanisms involved in several methods, such as radioimmunoassay or enzyme-linked immunosorbent assay (ELISA) assay, are relatively complex and not always fully clarified. More elaborate calibration models must be selected to obtain accurate result quantification. For example, Robison–Cox proposed early the 4-parameter logistic model (4PL) [4]. The 5-parameter logistic model also exists. Both models can be used to predict the OD of  $Y$  of the antigen-bound fraction as a function of the ligand concentration  $X$  for several ligand binding methods. But they can also be inverted to predict the concentration of an unknown sample from its instrumental response.

4-Parameter logistic model (4PL)

$$Y = a_2 + \frac{a_1 - a_2}{1 + \left(\frac{X}{a_3}\right)^{a_4}} + E \quad (2.34)$$

Inverse 4PL

$$Z = a_3 \left( \frac{a_1 - a_2}{Y - a_2} \right)^{\frac{1}{a_4}} \quad (2.35)$$

Although the 4PL model is empirical, the parameters can be interpreted in a way that is satisfactory to an analyst:

- $a_1$  the smallest response of  $Y$  when  $X$  tends to 0 and corresponds to the lower asymptote of the curve.
- $a_2$  the highest response of  $Y$  when  $X$  tends to infinity and corresponds to the plateau of the curve.
- $a_3$  the concentration of  $X$  when  $Y$  is halfway between its maximum and minimum.
- $a_4$  the curvature intensity.

The major difference between this model and the linear models is that it cannot be transformed into an additive polynomial as defined before. It is strictly nonlinear “in the parameters.” The theory of least-squares nonlinear regression will not be presented here but is well-documented and illustrated in several books [5]. The nonlinear regression algorithms, both unweighted and weighted, are mostly iterative and quite different from OLS and WLS applicable to the linear case. To demonstrate the feasibility of nonlinear regression is now straightforward and accessible using modern computing facilities.

A practical application to the 4-parameter logistic calibration function of Eq. (2.34) is presented. This is an example applied to a dataset collected with an ELISA method developed to determine an interleukin in blood. Because the number of coefficients to be estimated is substantial (4) and the model is more complex, it is recommended to collect more data than for a simple linear model. Data are gathered in Table 2.4, with the concentration  $X$  in pg/ml and the

**Table 2.4** ELISA<sup>a)</sup> – calibration for the determination of interleukin 6 using nonlinear regression.

| Concentration $X$ (pg/ml) | $Y_{i1}$ | $Y_{i2}$ | $Y_{i3}$ |
|---------------------------|----------|----------|----------|
| 3.91                      | 0.326    | 0.348    | 0.331    |
| 7.81                      | 0.361    | 0.387    | 0.366    |
| 15.63                     | 0.430    | 0.458    | 0.442    |
| 31.25                     | 0.571    | 0.593    | 0.582    |
| 62.50                     | 0.873    | 0.911    | 0.874    |
| 125.00                    | 1.380    | 1.402    | 1.419    |
| 250.00                    | 2.167    | 2.174    | 2.143    |
| 500.00                    | 2.756    | 2.820    | 2.732    |

Response unit is optical density.

a) Unpublished personal data.

calibrators is large  $I = 8$ , as well as the number of replicates  $J = 3$  per calibrator, i.e. altogether 24 calibration measurements are necessary.

The built-in function `LINEST` is not able to handle such a model. It could be possible to set up a template worksheet using the Excel add-in called Solver, but it requires some expertise. The simplest solution consists in developing a short Python script like Resource C. Returned results are stored in two structures, one called `pop` that contains the values of the coefficients: in this example  $a_1 = 0.33166$ ,  $a_2 = 3.51081$ ,  $a_3 = 202.923$ ,  $a_4 = 1.33290$ . The other called `cov` contains the covariance matrix. The interpretation of nonlinear regression parameters is quite different from OLS or WLS. It is important to check if the different coefficients are not correlated. A strong correlation between two coefficients would indicate that they are redundant and the model is overfitted. In this example all covariances indicate that the coefficients are independent.

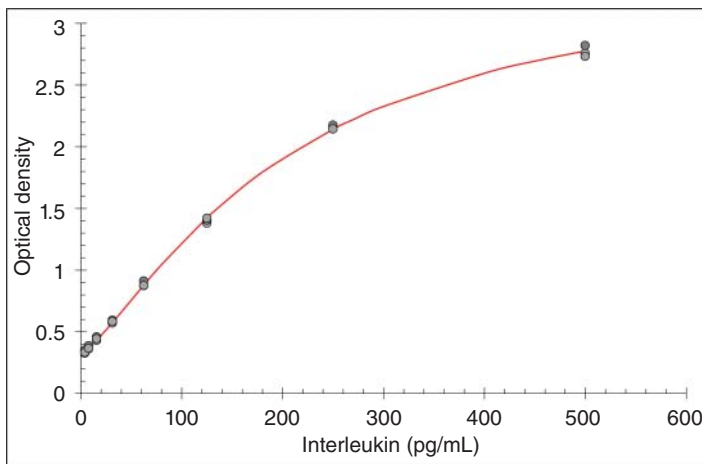
**Figure 2.4** ELISA – determination of interleukin 6. Nonlinear calibration curve and observed measurements.

Figure 2.4 illustrates the ELISA dataset and the continuous line the estimated calibration function. To draw this line, it is necessary to separately compute the predicted response for each concentration using Eq. (2.34). The following short worksheet is a possible application; the column headline “ $\hat{Y}$ ” stands for  $\hat{Y}$  and contains some predicted response values.

|    | A                                            | B               | C     | D     | E       | F                                       |
|----|----------------------------------------------|-----------------|-------|-------|---------|-----------------------------------------|
| 1  | Nonlinear regression for Interleukin (pg/ml) |                 |       |       |         |                                         |
| 2  | Coefficients                                 |                 |       | a1    | 0.33166 |                                         |
| 3  |                                              |                 |       | a2    | 3.51081 |                                         |
| 4  |                                              |                 |       | a3    | 202.923 |                                         |
| 5  |                                              | Optical density |       | a4    | 1.3329  |                                         |
| 6  | Concentration X (pg/mL)                      | Yi1             | Yi2   | Yi3   | Y^      |                                         |
| 7  | 3.91                                         | 0.326           | 0.348 | 0.331 | 0.348   | =E\$3+((E\$2-E\$3)/(1+(A7/E\$4)^E\$5))  |
| 8  | 7.81                                         | 0.361           | 0.387 | 0.366 | 0.372   | =E\$3+((E\$2-E\$3)/(1+(A8/E\$4)^E\$5))  |
| 9  | 15.63                                        | 0.430           | 0.458 | 0.442 | 0.433   | =E\$3+((E\$2-E\$3)/(1+(A9/E\$4)^E\$5))  |
| 10 | 31.25                                        | 0.571           | 0.593 | 0.582 | 0.574   | =E\$3+((E\$2-E\$3)/(1+(A10/E\$4)^E\$5)) |
| 11 | 62.5                                         | 0.873           | 0.911 | 0.874 | 0.879   | =E\$3+((E\$2-E\$3)/(1+(A11/E\$4)^E\$5)) |

### Resource C Nonlinear calibration (Python).

For this script, it is necessary to import the `curve_fit` function from the `scipy.optimize` package that performs the nonlinear regression.

```
from scipy.optimize import curve_fit
```

Create a subroutine called `LogisticModel` to declare the 4-parameter of the logistic model. It must be input before any other command line. Beware, the calibration formula is no longer described as a text variable as it was for OLS regression with Python.

```
def LogisticModel(x, a1, a2, a3, a4):
    return(a2 + ((a1 - a2)/(1 + (x / a3)**a4)))
```

Calibration data are stored in local variables but can be input from an external file.

```
X = [3.91,3.91,3.91,7.81,7.81,7.81,15.63,15.63,15.63,
31.25,31.25,31.25,62.5,62.5,62.5,125,125,125,250,250,250,
500,500,500]
Y = [0.326,0.348,0.331,0.361,0.387,0.366,0.43,0.458,
0.442,0.571,0.593,0.582,0.873,0.911,0.874,1.38,1.402,
1.419,2.167,2.174,2.143,2.756,2.82,2.732]
```

The application of the `curve_fit` function is direct. Other arguments can be added, such as initial values of the coefficients when the function may have several optimums. In this example, it is not useful.

```
pop, cov = curve_fit(LogisticModel, X, Y)
```

Print results

```
print(pop) # coefficients
print(cov) # covariance matrix
```